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Original Research Article

A methodological approach to the genetic identification of native Brook Trout (*Salvelinus fontinalis*) populations for conservation purposes

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ABSTRACT

As biodiversity is transformed on a global scale, identifying and protecting populations unaffected by habitat disturbances is essential to ensuring natural genetic diversity is not irretrievably lost. Furthermore, knowledge of the interactions between endemic populations and human-introduced conspecifics is imperative to the creation of management strategies that conserve and enhance population viability. In this study, these issues were examined for Brook Trout (*Salvelinus fontinalis*) populations from the Northeastern United States through the characterization of the spatial genetic structure of over 2000 wild fish, collected from 83 different sampling locations, and encompassing 18 individual watersheds. Population genetic analyses of microsatellite data at 13 loci demonstrate that, despite centuries-long stocking practices, roughly half of the populations examined exhibit genetic structure consistent with native origins. In addition, when examining the populations that revealed influence from stocked con-specifics, only a limited proportion of ancestry was attributed to stocked fish in the majority of cases. The methodological approach presented in this study for identifying native populations demonstrated strong population subdivision within and among populations, watersheds, and drainage basins. These findings are consistent with past inferences regarding native population structure of cold-water fish species and represent a detailed accounting of the remaining Brook Trout populations still intact in their native waters. Furthermore, the extent of such populations greatly exceeds what had been previously identified and prioritized as 'heritage' based on stocking history. The methods and data presented here may serve as an important tool to inform diversity maintenance, species recovery planning, and biodiversity conservation in this and other genetically unique and historically important native species.
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1. Introduction

Native species represent an important part of any given area's natural heritage, and also play an integral role in the trophic web supporting their ecological community, thereby enriching and stabilizing local ecosystems (Vila et al., 2011; Nystrom, 2017). In addition, a large number of studies that have examined evolutionary potential in native populations have demonstrated local adaptations, suggesting that many populations may be uniquely suited to their endemic habitats on both regional and broader scales (O'Neil et al., 2014; Colautti and Lau, 2015; May-McNally et al., 2015; Bible and Sanford, 2016). Thus, the identification of regional populations that have potentially been uninfluenced by human-mediated translocations should be a priority for environmental managers and conservationists looking to restore and conserve species diversity across the globe.

Brook Trout are a culturally and economically important fish in the Northeastern United States and were once abundant in the cold-water lakes, streams, and rivers throughout the region (Melstrom et al., 2015; Caputo et al., 2016). Nevertheless, two centuries of accelerated habitat disturbances in the form of deforestation, overfishing, the introduction of non-native competitors, acidic deposition, and habitat fragmentation have led to a sharp decline in viable habitat for this species (Caputo et al., 2016; Miller et al., 2016; Zastavniouk et al., 2017). In addition to historical anthropogenic factors influencing Brook Trout population structure in the region, changes in seasonal climate regimes are expected to outpace compensatory responses (Bassar et al., 2016; Poesch et al., 2016; Lynch et al., 2016), with estimates by the United States Environmental Protection Agency (EPA) based on future climate projections predicting a 50–100% decline in Brook Trout abundance by the year 2100 due to warming waters and reduced precipitation regimes (EPA, 2015). Thus, it is imperative that populations contributing to landscape-level genetic diversity be both identified and protected to preserve their adaptive potential and natural genetic constitution (Marty et al., 2015; Shafer et al., 2015).

Compounding concerns associated with habitat degradation is an extended history of supplemental stocking throughout the region dating back two centuries (Daniels, 2011). Supplemental stocking practices were initially established to enhance fishery resources, while in turn providing greater angling opportunity. A wide range of research over the past several decades, however, has demonstrated the potential negative impacts of such practices (Einum and Flemming, 2001; North et al., 2006; Le Cam et al., 2015). Potential negative effects of stocking into wild populations include decreases in effective population size and genetic diversity (Eldridge et al., 2007; Laikre et al., 2010), the loss of local adaptation due to outbreeding depression (Gharrett et al., 1991; Goldberg et al., 2005), and genetic homogenization across the landscape (Lamaze et al., 2012; Valencia et al., 2017).

Past studies that have looked specifically at hatchery strains used for supplementation in New York State have reported comparatively reduced growth, survival, longevity, and reproductive success in hatchery-domesticated fish compared to wild-reproducing fish sampled from the same waters (Webster and Flick, 1981; Lachance and Magnan, 1990). Given the extended history of stocking in the Northeast, state officials in New York State have devoted great effort to protecting a small number of populations that are believed to be native in origin and genetically pristine, based on historical documentation (Keller, 1979). These populations are referred to as "heritage" strains by the New York State Department of Environmental Conservation (NYSDEC, 2017), yet the genetic origins of the vast majority of wild Brook Trout populations found throughout New York State and the Northeastern United States remain unknown.

A wide range of studies have demonstrated the utility of employing Bayesian clustering algorithms to measure the level of introgression from stocked fish into wild populations (Sanz et al., 2009; Lamaze et al., 2012; Glover et al., 2013; Scribner et al., 2015; Bruce and Wright, 2018; White et al., 2018), although these analyses are dependent on the inclusion of reference samples from the supplemented populations. In the Northeastern United States, many populations historically used for supplementation have been lost and, in the majority of cases, historical samples are not readily available. In addition, New York State is home to a wide range of regional and private hatcheries, making the acquisition of reference samples for all historically and contemporary supplemented populations infeasible; therefore, determining the genetic origin of wild sampled populations in the region is highly precarious. Nevertheless, literature available on the origin of strains historically used for supplementation in New York suggests that the majority of these populations were sourced from non-local waterbodies or have origins consistent with domestication in a hatchery setting (Vincent, 1960; Moyle, 1969; Webster and Flick, 1981, Perkins et al., 1993a, 1993b). In either case, we would expect their genetic structure to be inconsistent with natural patterns exhibited in indigenous populations found throughout the region.

Previous examinations of biogeographic structure in native Brook Trout, and other cold-water fish species, have indicated strong differentiation among both regional watersheds and larger drainage basins based on postglacial dispersal and subsequent recolonization patterns (Perkins et al., 1993a, 1993b; Danzmann et al., 1998; Castric et al., 2001). Given the extended history of stocking in the region, two alternative hypotheses can be extended regarding extant Brook Trout biogeographic structure as characterized by patterns of genetic differentiation within and between watersheds. The first is that widespread stocking has either homogenized or otherwise altered the level of diversity previously present in the region, and Brook Trout genetic structure is therefore no longer consistent with previously inferred biogeographic patterns. The other is that stocking has had a limited impact on many populations, given that native fish may hold a selective advantage relative to non-local or hatchery-reared conspecifics, whether that advantage takes the form of better tolerance to local environmental variables or greater competitive abilities related to food, mates, territory, or some combination thereof. Strong population genetic subdivision between watershed and drainage basin boundaries would be consistent with limited influence from stocked fish, in support of the latter hypothesis, while weak or even arbitrary differentiation would suggest that an extended history of

stocking has permanently altered Brook Trout genetic structure across the landscape, and the hypothesis of a selective advantage can be rejected.

In this study, we present a systematic approach for identifying populations that have been potentially uninfluenced by decades-long stocking activities using well-established techniques for genetic analysis, even when reference samples for historically stocked strains are potentially unavailable. To achieve this goal, we first compared each sampled population with currently stocked strains to determine the proportion of admixture attributed to conspecifics across the landscape, based on contemporary management practices dating back to 1980 (Keller, 1979). We then compared populations that exhibited minimal signs of introgression from stocked fish to each other and to previously documented native populations, using various measures of genetic differentiation at different hierarchical levels. Finally, we integrate these findings into a framework based on postglacial recolonization routes and watershed geography to determine if the genetic structure of these populations reflects previously inferred patterns of native fish biogeography in the Northeastern United States. We then conclude this study by discussing how such methods may more broadly aid in the identification of other genetically unique and important populations of conservation concern.

2. Methods

2.1. Study sites and sample collection

Tissue samples and genomic DNA from wild-collected Brook Trout sampled across 18 watersheds, spanning over 550 km^2, were included in this study (n = 2193). All sample collections took place between May 1 of 2013 and September 1 of 2017. The majority of Brook Trout sampling was carried out using a backpack electrofishing unit, and sample sites were chosen based on accessibility to State lands, past survey records, and documented habitat suitability across the region (EBTJV, 2018). Several sites, including Bay Pond, Falls Pond, Ausable Lake, and the Sagamore watershed were sampled using traditional angling techniques or trap nets. All other fish were generally collected from 50 to 100 m stream reaches, using anywhere from one to four successive shocking passes with one to three fish netters present. All tissue samples were taken using sterile technique and stored in 95% ethanol (EtOH). Seven additional reference groups were also included in this study, acquired between 2015 and 2016 from two state-administered hatcheries (n = 185) and from the Cold Spring Harbor hatchery (n = 24). These populations were specifically bred for stocking purposes and populations from the state-administered hatcheries are the only Brook Trout strains commonly used for supplementation on public lands in New York State, while the Cold Spring Harbor hatchery raises Brook Trout that were harvested from, and are specifically stocked on the Long Island peninsula. Out of the 7 stocked strains compared in this study, 4 are used exclusively to enhance angling opportunity and biodiversity in the Adirondack Park. New York is also home to a number of private and commercial hatcheries (NYDEC, 2017), for which genetic samples were not available. Nevertheless, the vast majority of sampled sites were on public land and unlikely to see influence from stocking implemented by private landowners elsewhere. In addition, at least some of the Brook Trout strains stocked by private hatcheries in New York were derived from the same State strains examined in this study.

2.2. DNA extraction and genotyping

The majority of DNA extractions for this study (n = 1953) were carried out following the tissue protocol included with the QIAGEN DNeasy tissue kit (QIAGEN, Inc., Valencia, California), with the exception of samples from Western New York (n = 240) that were extracted by the United States Fish and Wildlife Service using the KingFisher Flex Magnetic Particle Processor (ThermoFisher Scientific) and the Puregene (Gentra Systems, Inc.) DNA extraction methods, following manufacturers' protocols. Western New York sample sites were chosen based on preliminary genetic data from another study that suggested a high likelihood of native origin based on genetic structure analysis that examined introgression from locally-stocked fish (Beer et al., in prep).

Thirteen autosomal microsatellite loci were genotyped using the same procedures for all individuals examined in this study. Polymerase Chain Reaction (PCR) was used to amplify loci using thirteen primer pairs created specifically for Brook Trout (King et al., 2012). The forward primers were fluorescently labeled with HEX, FAM, or NED dye for downstream electropherogram analysis. PCR-related methods resulted in five 20 μl multiplex PCR reactions and one single 20 μl PCR reaction for each individual. PCR amplification was carried out using two Bio-Rad T100 thermal cyclers. Fragment analysis using an internal size standard (Liz600, Applied Biosystems) was performed using an Applied BioSystems 3730XL DNA Analyzer. The automated scoring of genotypes was carried out using the Geneious 4.0 (Biomatters Ltd.) software package. All automated genotype calls were confirmed by eye.

2.3. Initial neutrality testing and admixture analysis

MICRO-CHECKER was first used to identify any potential null alleles and/or large allelic drop outs present in the data set (Van Oosterhout et al., 2004). COLONY was then used to identify full sibling families at each sampling location. Full siblings were removed until a single-family representative from each pair remained at the population level, to ensure the genetic structure was not masked by a few dominant family groups (Jones and Wang, 2010). All thirteen microsatellite loci for genotyped fish were then subjected to outlier tests of neutrality, which were run using the LOSITAN workbench (Antao et al.,

2008) assuming 83 populations, based on individual sampling locations and the seven stocked strains. One million simulations were applied under a drift with migration model, while assuming a stepwise mutation model for each sampled location. The initial simulation run was used to remove potentially non-neutral loci before computing the genomic mean F_{ST} for further simulations, and a bisection algorithm over repeated simulations was subsequently used to approximate a desired F_{ST}.

To estimate the level of admixture attributed to stocked strains across sampling locations, a Bayesian clustering analysis was performed using the program STRUCTURE 2.3.4 (Pritchard et al., 2000). STRUCTURE has been shown to outperform other programs for hybrid analysis, and has been used to accurately estimate introgression from stocked conspecifics in this species when examined using both wild-collected fish and simulations, employing the same microsatellites used in this study (Marie et al., 2011). Each sampled location was run individually with the six strains currently and historically used for stocking across New York State by the NYSDEC, to prevent potential complications in estimating admixture across a large number of putative populations of unequal sample size (Puechmaille, 2016; Wang, 2017). Sampled locations from Long Island were additionally run with the Cold Spring Harbor hatchery strain, which was originally sourced from the region and is solely used for supplementation on Long Island. Model parameters included a burn-in step of 250,000 iterations followed by 500,000 Markov Chain Monte Carlo (MCMC) iterations, using an admixture model. The initial alpha value was set to 0.20 ($ALPHA = 1/\text{putative K}$), assuming uncorrelated allele frequencies (Wang, 2017). Under this model, individuals are the unit of analysis and their multilocus genotype can be inferred as admixed from any number of K source populations. We performed ten runs each for $K = 4$ through $K = 8$ on all sampling locations, with the exception of the Long Island samples, for which we performed ten runs each for $K = 5$ through $K = 9$ to account for the additional Cold Spring Harbor genotypes.

Runs were examined for potential K-values, assuming the following four possible scenarios: 1) the genotypes of a sampled location corresponded entirely to the stocked strains; 2) the genotypes of a sampled location are admixed and correspond partially to stocked strains, and partially to a population for which reference genotypes are not available; 3) the genotypes of a sampled location correspond entirely to a population for which reference genotypes are not available; or 4) any of scenarios 1–3, with additional sub-structuring across the sampled location. The web program CLUMPPAK (Kopelman et al., 2015) was then used to combine Q-value results from independent runs, and STRUCTURE HARVESTER (Earl, 2012) was used to calculate the best-supported K-value determined by the mean likelihood Ln(K) method (Pritchard, 2000; Wang, 2017). Mean estimated admixture from stocked fish (q-membership; Q_M) was then reported and visualized using bar plots, executed in the R package ggplot2 (Wickham, 2009). Finally, a threshold value was determined based on the admixture estimates associated with stocked strains exhibited by the previously identified and/or confirmed native populations included in this study, which was subsequently used to exclude potentially introgressed populations from all downstream analysis (proportion of mean ancestry associated with stocked strains at levels greater than 5%; $Q_M > 0.05$).

2.4. Summary statistics and hierarchal genetic structure of putatively native strains

All putatively native sampling locations ($Q_M > 0.05$) identified in the STRUCTURE analysis were subjected to tests for deviations from Hardy-Weinberg equilibrium (Guo and Thompson, 1992) and for linkage disequilibrium, using an exact test in ARLEQUIN 3.5 (Excoffier and Lischer, 2010) and the log likelihood ratio test (G-test) in GENEPOP 1.2 (Rousset, 2008), respectively. Summary statistics were also calculated, including observed and expected heterozygosity in ARLEQUIN, and the inbreeding coefficient F_{IS} (Weir and Cockerham, 1984), which was computed across all loci for each sampling location using GENEPOP. Allelic richness was also estimated, using the rarefaction method to correct for differences in sample size with FSTAT 2.9.3.2 (Goudet, 2001). Finally, effective population size (N_e) estimates, with 95% parametric confidence intervals, were calculated using NeEstimator V2.01 (Do et al., 2014).

To examine the comparative hierarchical genetic structure of putatively native fish, a Bayesian Clustering algorithm was again implemented with the program STRUCTURE. All remaining sampled locations ($Q_M < 0.05$) were run simultaneously to assess the number of populations at each hierarchical level within the dataset based on watershed boundaries and the ΔK statistic (Evanno et al., 2005). ΔK is based on the rate of change in the log probability of data between successive K-values, and thus has been shown to accurately identify the uppermost hierarchical level of genetic structure across multiple scenarios tested (Evanno et al., 2005).

Model parameters included a burn-in step of 250,000 iterations followed by 500,000 Markov Chain Monte Carlo (MCMC) iterations, using an admixture model. The initial alpha value was this time set to 0.025 ($ALPHA = 1/\text{putative K}$), assuming uncorrelated allele frequencies (Wang, 2017). We performed ten runs each for $K = 18$ (the number of watersheds) through $K = 43$ (the number of sampling locations, plus 3) to account for all possible population designations in-between. STRUCTURE HARVESTER was then used to calculate the uppermost level of hierarchical structure (ΔK statistic). Sample sites that clustered together were then split out from the dataset and run again employing the same methodology described above, but adjusting the number of K-values tested to reflect the number of sampling locations examined.

In addition to cluster analysis, pairwise genetic distance measures (D) were calculated using the GENALEX 6 software package (Nei, 1978; Peakall and Smouse, 2006) and converted into tree from with the R package PEGAS (Paradis, 2010). The tree was rooted with the Twomile population and then arranged in decreasing node order using Figtree (Rambaut, 2009). The Twomile population is a previously confirmed 'heritage' population in the Allegheny basin, the only inland region of the state that was not previously covered by the Laurentide ice sheet and therefore one of the first areas likely to be colonized by Brook Trout in the state (Dyke and Prest, 1987, Perkins et al., 1993a, 1993b). Pairwise F_{ST} tests (Hardy Weinberg Equilibrium not

assumed) were carried out applying Weir and Cockerham's unbiased estimator of F_{ST} (theta) using FSTAT. Resulting F_{ST} values were then plotted against Euclidian distance to test for a pattern of isolation (Rousset, 1997), and mean F_{ST} values were displayed using boxplots generated with the R package ADEGENET (Jombart, 2008).

2.5. Analysis in the context of previously inferred biogeographic structure

To test for patterns of biogeographic structure based on previous inferences related to postglacial recolonization, watershed boundaries, and drainage basins (Perkins et al., 1993a, 1993b), we employed a discriminant analysis of principal components (DAPC) in the R package ADEGENET (Jombart, 2010). DAPC is a multivariate analysis method designed to identify and describe clusters of genetically-related individuals. We used a sequential K-means and model selection approach to infer the number of genetic clusters within the putatively native data set. This approach allows for the extraction and visualization of between-population differentiation at multiple hierarchical levels (Jombart, 2010). Results of the DAPC analysis were plotted at the level of sampling location, watershed, and drainage basin to observe differences attributed to hierarchal population structure across individuals and the landscape. In addition to DAPC analysis, we performed an analysis of molecular variance (AMOVA) using GENALEX, to measure differences within, among, and between sample groupings at each previously described hierarchical level.

3. Results

3.1. Admixture across sampled locations

MICRO-CHECKER detected no null alleles or allelic drop out within the complete dataset. COLONY identified 20 full siblings across all sampled locations, therefore one individual from each pair was removed from further analysis. The program LOSITAN determined all loci to fall in the range of neutrality. Results of neutrality tests are reported in Supplemental 1. Results of the initial introgression analysis carried out with the program STRUCTURE are reported in Table 1 and Fig. 1. The proportion of mean admixture attributed to stocked strains varied widely among sampling locations. Four sampling locations from the Adirondacks and two sampling locations from Long Island exhibited admixture estimates attributed entirely to stocked fish ($Q_M = 1.0$). Only two additional populations exhibited admixture estimates associated with stocked fish in the upper range of 50–99% ($Q_M = 0.5–0.99$; Blue Ledge and Huntley Inlet). The majority of fish from the Adirondacks exhibited admixture estimates attributed to stocked fish in the 5–30% range ($Q_M = 0.05–0.30$), while the majority of fish from the Catskills and Eastern New York fell in the low range 0–5% ($Q_M = 0.00–0.05$). Western New York showed the lowest level of admixture estimates attributed to stocked fish out of all of the areas examined in the study, although these sampling locations were prescreened in a preliminary study and therefore should not be considered representative of the region as a whole (Beer et al., in prep).

All of the previously confirmed native or heritage strains fell into the lower range of the populations examined, producing admixture estimates associated with stocked fish at less than 5% ($Q_M < 0.05$), including their brood strains. No population produced an admixture estimate attributed to stocked fish at 0% ($Q_M < 0.00$), however, when compared to strains used for supplemental stocking. Given that all previously confirmed native or heritage strains exhibited ancestry estimates attributed to stocked fish at levels less than 5% ($Q_M = 0.05$), this value was used as the threshold in determining significant introgression from contemporary stocked fish. Examining the dataset as a whole, 40 populations (985 fish; 50% of fish examined in this study) exhibited introgression estimates from stocked fish at levels < 5%, while 27 populations (703 fish; 36% of fish examined in this study) exhibited introgression at levels < 30%, and 16 populations (286 fish; 14% of fish examined in this study) exhibited introgression estimates at in the range of 30%–100%.

3.2. Diversity indices and genetic structure in putatively native populations

Tests for Hardy-Weinberg equilibrium identified 6 loci that exhibited significant departures from equilibrium out of 520 comparisons (following sequential Bonferroni correction of $\alpha = 0.05$; initial nominal P-value = 0.00009). These loci included: locus *SfoC86* in Spencer, locus *SfoD100* in the Tunis Lake brood strain, locus *SfoC38* and locus *SfoC115* in Newton Run, locus *SfoC115* in the Swan River, and locus *SfoC115* in Birch Brook. Tests for linkage disequilibrium across all pairs of loci produced only 7 significant comparisons out of 2944 (following sequential Bonferroni correction of $\alpha = 0.05$; initial nominal P-value = 0.00002). Summary statistics across all putatively native sampling locations are reported in Table 2. Expected heterozygosity varied somewhat between sampling locations (mean $H_E = 0.555$), but generally fell in the range of 0.405–0.705. L. Brownell Trib from the Upper Allegheny watershed exhibited the highest expected heterozygosity ($H_E = 0.703$), while Site 316384 from the Rondout watershed produced the lowest ($H_E = 0.403$). Measures of allelic richness across all sampled locations showed considerably more variation than expected heterozygosity. Mean allelic richness across all putative populations was measured at 45.95. As with expected heterozygosity, L. Brownell Trib from the Upper Allegheny watershed exhibited the highest measure ($A = 65.76$), while Site 316384 produced the lowest ($A = 23.36$). Wright's inbreeding coefficient (F_{IS}) calculated across all samples showed only minor variations, while mean F_{IS} across all samples was calculated at −0.02. None of the sampling locations exhibited a heterozygote excess, while only two sampling locations exhibited deficiencies: Crow in the Niagara watershed and Lewis Brook in the Hudson-Hoosic watershed. Estimates of effective population size

Table 1
Sample sizes and locations for all individuals included in the study, ordered by introgression estimates. N=Number of specimens, Q_M = admixture proportion associated with stocked strains (q-membership; 0–1.0) with standard deviation as estimated by STRUCTURE.

Sampling location	N	Q_M	Sampling location	N	Q_M
Windfall (brood)*	31	0.001 (0.001)	Site 35008	34	0.061 (0.035)
Horn (brood)*	31	0.001 (0.001)	Vanderwhacker 1	12	0.065 (0.047)
L. Tupper (brood)*	31	0.002 (0.001)	Neversink East	30	0.065 (0.042)
Site 316384	24	0.004 (0.001)	Spring	30	0.067 (0.030)
Spencer	30	0.004 (0.002)	Carmans	5	0.071 (0.014)
Johnson Trib	22	0.005 (0.001)	Platt	29	0.077 (0.079)
Tunis (brood)*	29	0.005 (0.002)	Site 21013	30	0.077 (0.072)
Newton Run	30	0.006 (0.005)	Site 23003	26	0.097 (0.045)
Crow	28	0.006 (0.002)	Shanty Bottom	27	0.097 (0.050)
Tuthills	16	0.006 (0.003)	Vanderwhacker 2	30	0.103 (0.042)
Russell*	30	0.006 (0.002)	Site 21005	23	0.106 (0.038)
Canacadea 1	29	0.007 (0.006)	Poestenkill	22	0.116 (0.036)
Biscuit	30	0.008 (0.009)	Site 7028	26	0.127 (0.135)
Twomile*	27	0.008 (0.021)	Fly Outlet	30	0.158 (0.235)
Canacadea 2	16	0.009 (0.004)	Ausable Lake	60	0.164 (0.178)
Yaphank	25	0.001 (0.005)	Durgin	27	0.179 (0.048)
Gulf	24	0.011 (0.005)	Site 22004	32	0.192 (0.045)
Redwater	30	0.011 (0.005)	Site 30003	26	0.202 (0.030)
Kinne	27	0.011 (0.005)	Frontier	13	0.204 (0.122)
Site 316269	28	0.012 (0.023)	Bear Hole Brook	9	0.206 (0.367)
Site 316395	28	0.012 (0.010)	Delaware East	9	0.214 (0.111)
Lewis*	30	0.015 (0.025)	Sagamore	42	0.217 (0.103)
Casey Inlet	15	0.015 (0.005)	Big Brown	30	0.219 (0.086)
Connetquot*	22	0.016 (0.007)	Moss Inlet	29	0.264 (0.345)
Upper Canaan	12	0.016 (0.002)	Site 31007	30	0.271 (0.104)
L. Charley*	29	0.016 (0.010)	Nate	26	0.314 (0.056)
Neversink West	32	0.017 (0.014)	Patchogue	9	0.327 (0.132)
Site 316164	20	0.021 (0.005)	Site 7003	19	0.392 (0.112)
Nissequogue	30	0.021 (0.011)	Site 28018	30	0.406 (0.056)
Honnedaga*	30	0.021 (0.035)	7th Inlet	28	0.428 (0.040)
L. Brownell Trib	28	0.022 (0.016)	Site 22017	14	0.440 (0.048)
Snyder	28	0.027 (0.008)	Windfall Outlet	23	0.443 (0.103)
Buck	19	0.032 (0.031)	Birch	29	0.499 (0.084)
Swan	10	0.032 (0.006)	Huntley Inlet	30	0.753 (0.038)
Birch Brook	13	0.033 (0.010)	Blue Ledge	16	0.984 (0.001)
Dix-Elk*	25	0.040 (0.025)	Chain Inlet	4	1.000 (0.000)
Site 29012	19	0.041 (0.023)	Niagara	3	1.000 (0.000)
Slide	17	0.045 (0.014)	Massapequa	2	1.000 (0.000)
Calamity	29	0.047 (0.015)	Shu Swamp	30	1.000 (0.000)
Trout Brook	11	0.048 (0.025)	Bay	17	1.000 (0.000)
B. Bear	28	0.051 (0.021)	Falls	6	1.000 (0.000)
Site 316327	14	0.052 (0.018)			

across locations varied widely, and many locations produced estimates with an infinite upper confidence interval. Effective population size (N_e) averaged 73.92 across all locations, with Buck Creek exhibiting the lowest value ($N_e = 7.1$) and site Canacadea 2 having the highest by a substantial margin ($N_e = 917.9$).

The initial best-supported K-value for the hierarchical STRUCTURE analysis showed strong support for 26 populations ($\Delta K = 26$) out of the 40 sampling locations examined (Supplemental 2). While many sampling locations break out as their own individual cluster at $K = 26$, the majority of sample sites that do not are found almost exclusively within the same watershed (Fig. 2). The exceptions to this rule are Windfall and L. Tupper (both brood ponds derived from the larger St. Lawrence drainage basin) and Buck, which was originally sampled from the Black River watershed, but whose ancestry is highly admixed and partially attributed to two disparate populations; Lewis, from the Hudson-Hoosic watershed and Nissequogue, from the Northern Long Island watershed. Subsequent hierarchical STRUCTURE analysis shows all sampling locations eventually breaking out on their own with some partial ambiguity/admixture between them, the exception being the two pairs Canacadea 1 and Canacadea 2 and Snyder and Slide, both of which were determined to belong to the same cluster when run in tandem.

Pairwise genetic distance measures calculated in GENALEX are reported in Supplemental 3. When these measures are viewed in the form of a phylogram (Fig. 3), we see that distance measures reflect the proximity of sampling locations in most instances, and largely reflect the population groupings seen in the STRUCTURE plot. Sampled locations from the same drainage basin largely group together in all cases. One notable exception is again Buck, which groups more closely with the Housatonic watershed than the other Black River watershed sampling locations. Honnedaga shows strong differentiation from all other sampling locations. Many of the Lower Hudson River and Delaware River samples group tightly, while sampling locations in the Allegheny and Genesee River basins show more pronounced differentiation, but nonetheless group together. Interestingly, Lewis Hollow, which is technically in the Upper Hudson River drainage but connects to the Hudson River below

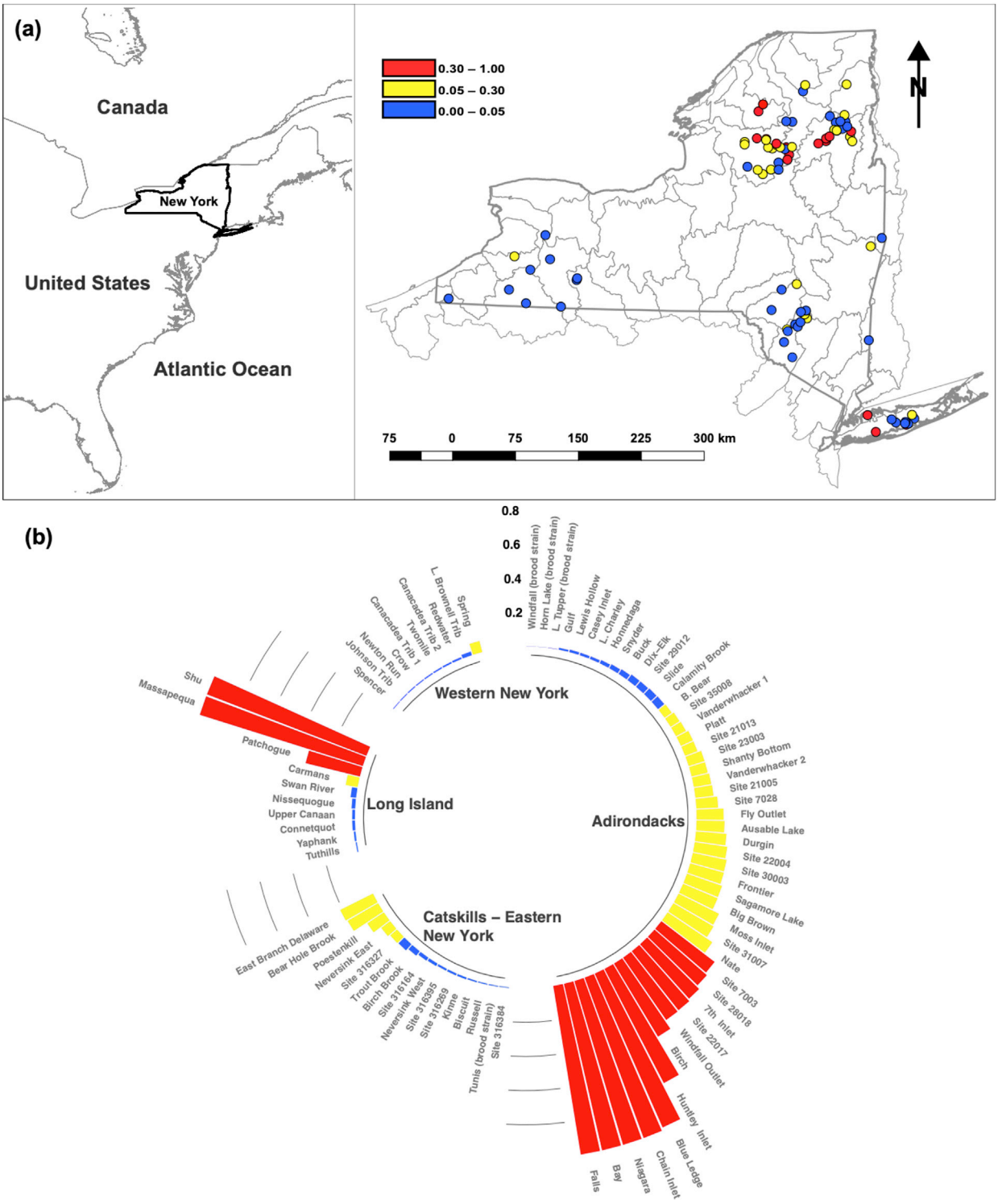

Fig. 1. Map of New York State, USA, depicting watershed boundaries and sampling locations (a), with corresponding circular bar plot (b). Sampling locations in the map, and bar plot, are shaded according to the proportion of admixture attributed to stocked fish at each sampling location (ranging between 0.0 and 0.05, 0.05–0.30, and 0.30–1.00), as estimated using the program STRUCTURE. Larger geographic regions, and mean admixture proportions associated with stocked fish are demarcated in the circular bar plot in black, while individual sampling locations are labeled in gray.

Table 2
Sample sizes and summary statistics for all sampling locations that exhibited less than 5% admixture from stocked strains. N = Number of specimens, H_E = Mean expected heterozygosity, H_O = Mean observed heterozygosity, A = Total allelic richness (based on a minimum sample size of 10 individuals), F_{IS} = Wright's inbreeding coefficient, N_e = Effective population size estimates, with 95% confidence intervals.

Watershed	Sampling location	N	H_E	H_O	A	F_{IS}	N_e
Upper Allegheny	Twomile	27	0.486	0.506	39.15	−0.035	42.3 (19.9–280.4)
	Newton Run	30	0.421	0.459	35.84	−0.092	35.2 (18.6–106.7)
	L. Brownell Trib	28	0.703	0.724	65.76	−0.032	44.9 (29.7–81.0)
	Johnson Trib	22	0.573	0.609	37.52	−0.066	7.7 (3.9–13.4)
Upper Genesee	Spencer	30	0.572	0.605	42.72	−0.059	94.0 (36.4 − ∞)
	Redwater	30	0.675	0.668	58.63	0.010	26.1 (18.9–38.6)
Housatonic	Site 316269	28	0.608	0.615	45.56	−0.014	19.0 (12.5–31.9)
Niagara	Crow	28	0.629	0.537	47.11	0.142[a]	29.3 (17.6–61.8)
Rondout	Birch Brook	13	0.533	0.503	48.99	0.057	39.9 (13.4 − ∞)
	Site 316164	20	0.507	0.500	40.52	0.014	12.7 (7.2–25.1)
	Site 316384	24	0.403	0.439	23.36	−0.133	–
Mohawk	Honnedaga	30	0.595	0.585	51.22	0.018	64.9 (33.6–284.6)
Tioga	Canacadea 1	29	0.491	0.503	36.29	−0.024	64.5 (22.7 − ∞)
	Canacadea 2	16	0.441	0.476	31.25	−0.086	917.9 (14.1 − ∞)
Black	Buck	19	0.615	0.623	47.29	−0.015	6.3 (3.3–9.8)
	Site 29012	19	0.647	0.623	60.68	0.037	102.8 (38.6 − ∞)
	Horn	31	0.599	0.638	51.63	−0.066	127.4 (51.8 − ∞)
St. Regis	Windfall	31	0.648	0.685	57.32	−0.059	39.4 (27.2–64.1)
Raquette	L. Tupper	31	0.596	0.619	50.27	−0.040	39.0 (25.6–69.2)
	L. Charley	29	0.619	0.606	48.45	0.021	19.5 (13.4–30.3)
Upper Hudson	Snyder	28	0.552	0.507	54.06	0.083	33.0 (21.6–58.8)
	Slide	17	0.514	0.505	51.52	0.019	75.6 (27.2 − ∞)
	Gulf	24	0.557	0.534	41.33	0.043	74.6 (26.3 − ∞)
	Dix-Elk	25	0.603	0.580	55.74	0.039	254.3 (67.3 − ∞)
	Calamity	29	0.634	0.636	54.92	−0.004	27.2 (19.0–42.4)
Ausable	Casey Inlet	15	0.467	0.451	45.95	0.035	–
Upper Delaware	Tunis	29	0.452	0.468	32.97	−0.038	–
Middle Delaware -	Kinne	27	0.587	0.603	45.72	−0.029	37.9 (20.7–110.00)
Mongaup - Brodhead	Site 316395	28	0.580	0.646	41.80	−0.116	76.8 (29.7 − ∞)
	Trout Brook	11	0.636	0.699	55.44	−0.104	–
	Biscuit	30	0.553	0.572	50.01	−0.034	23.2 (16.3–35.5)
	Neversink West	32	0.633	0.619	56.78	0.023	79.3 (45.7–218.0)
East Branch Delaware	Russell	30	0.435	0.431	39.32	0.010	23.3 (14.2–45.0)
Hudson - Hoosic	Lewis	30	0.507	0.456	38.82	0.102[a]	10.0 (6.8–14.7)
Northern Long Island	Nissequogue	30	0.558	0.576	48.41	−0.031	24.1 (16.3–39.2)
Southern Long Island	Yaphank	25	0.608	0.636	46.56	−0.045	21.0 (13.6–36.5)
	Swan	10	0.572	0.615	49.86	−0.077	26.1 (8.6 − ∞)
	Upper Canaan	12	0.479	0.513	35.74	−0.100	18.1 (4.9 − ∞)
	Tuthills	16	0.410	0.396	29.39	0.036	97.9 (11.1 − ∞)
	Connetquot	22	0.502	0.460	44.12	0.072	26.0 (14.5–66.5)

[a] Significant heterozygote deficit/excess at $P < 0.05$ after Bonferroni correction ($\alpha = 0.001$).

the Federal Dam in Troy, groups strongly with the Nissequogue sampling location, the only Long Island location that drains North into the Long Island Sound. The test for isolation by distance was highly significant, describing roughly 40% of the genetic variation seen among sampling locations as measured by F_{ST} (Fig. 3; F_{ST} data not shown). All F_{ST} comparisons were significantly different ($P < 0.05$) and, when viewed as box plots, sampling locations generally fell within the same range, with several minor exceptions (Supplemental 4). Most notably, Site 316384 and Tuthills showed heightened F_{ST} measures compared to all other locations, while the majority of other comparisons exhibited values in the 0.1 to 0.3 ranges.

3.3. Incorporating genetic analysis with landscape structure

The results of the DAPC analysis are shown in Fig. 4. When using the find. clusters function and implementing a Bayesian Information Criterion (BIC), ADGENET identified ~30 populations and inferred population designations for all individuals that were largely consistent with those produced by STRUCTURE (Supplemental 5). The DAPC ordination plots demonstrate that patterns of genetic structure strongly reflect watershed geography at all hierarchal levels and, in fact, the plots largely resemble the physical geography of the sampled range. Western New York samples are distributed more widely across the plot, suggesting heightened genetic diversity between those sampling locations. In contrast, the Long Island samples group tightly together, and appear more closely related to the Lewis Hollow and the Mohawk samples. Samples from the Delaware River watershed also group tightly and are slightly more differentiated from the rest of the region, especially given the inclusion of the Tunis brood strain. The Adirondack samples group tightly in the top right portion of the plot, and are highly differentiated from both the Western New York samples and Southern New York samples. Overall, the patterns of genetic structure exhibited in the DAPC analysis largely reflect those shown in the STRUCTURE analysis and phylogram.

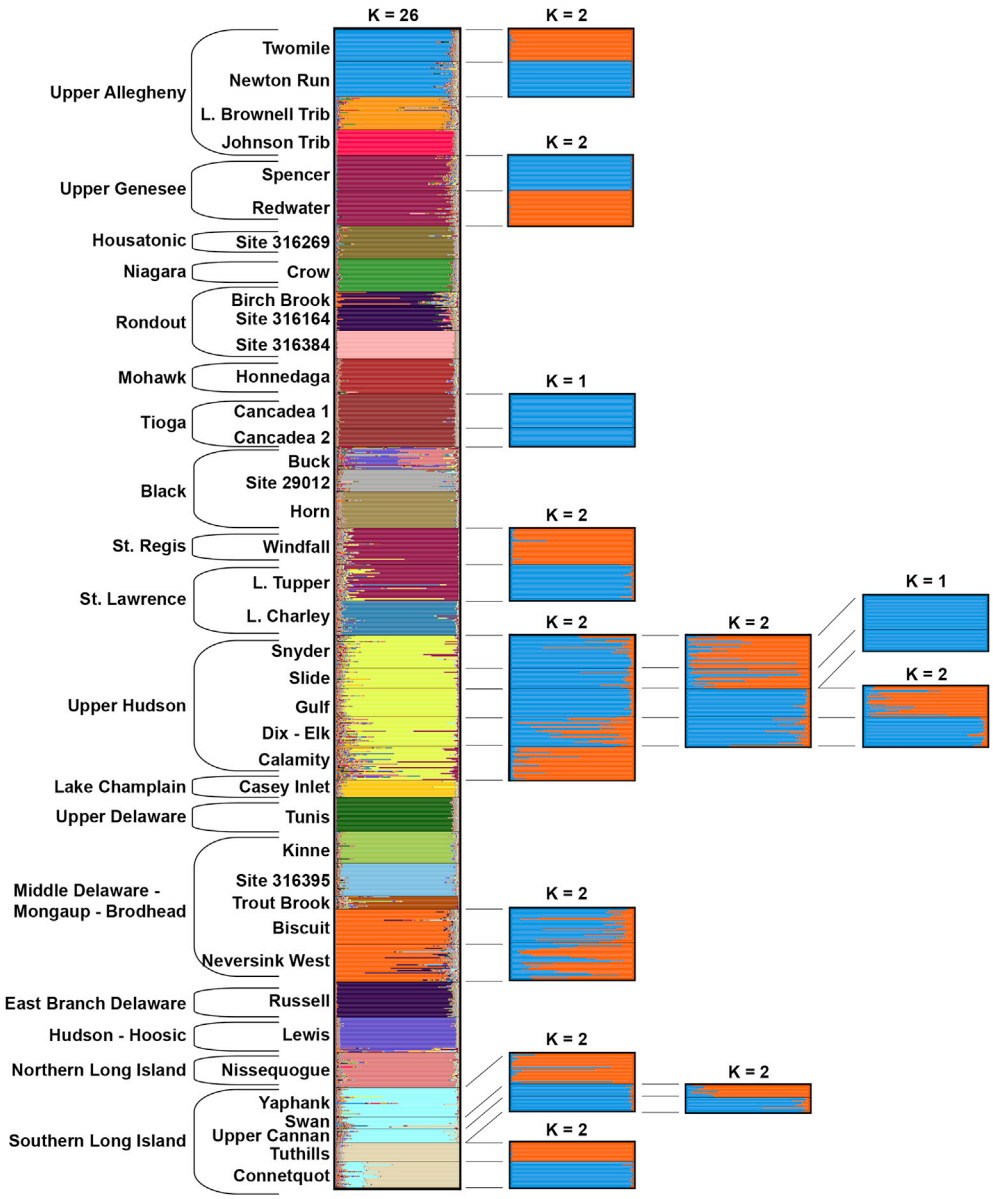

Fig. 2. Hierarchical STRUCTURE analysis for all fish included in the study for which initial stocked admixture estimates were measured at proportions of less than 5% (for combined ancestry attributed to any of 7 strains of fish commonly stocked in New York State since 1980). Individual sampling locations are labeled to the left of the bar plots, and encapsulated by labels corresponding to their larger watershed. Each horizontal line represents an individual fish, and colors represent their inferred ancestry from K ancestral populations at each hierarchical level. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

Results of the AMOVA analysis are reported in Table 3. When comparing Brook Trout at the population level, 29% of the observed variation was attributed to differences among populations, while 0% was attributed to among individual differences, and 71% was attributed to within individual differences. When comparing Brook Trout at the watershed level, 21% of the observed variation was attributed to watershed differences, 9% was attributed to among individual differences, and 70% was attributed to within individual differences. Finally, at the major drainage level, 18% of genetic variation was attributed to differences in the major drainage basins, 12% to among individual differences, and 70% to within individual differences.

4. Discussion

Understanding how historical and contemporary anthropogenic translocations have transformed native population structure is essential for making informed decisions related to conservation planning, species recovery, and diversity maintenance. The large-scale undocumented release of plants and animals across the globe has greatly altered native species

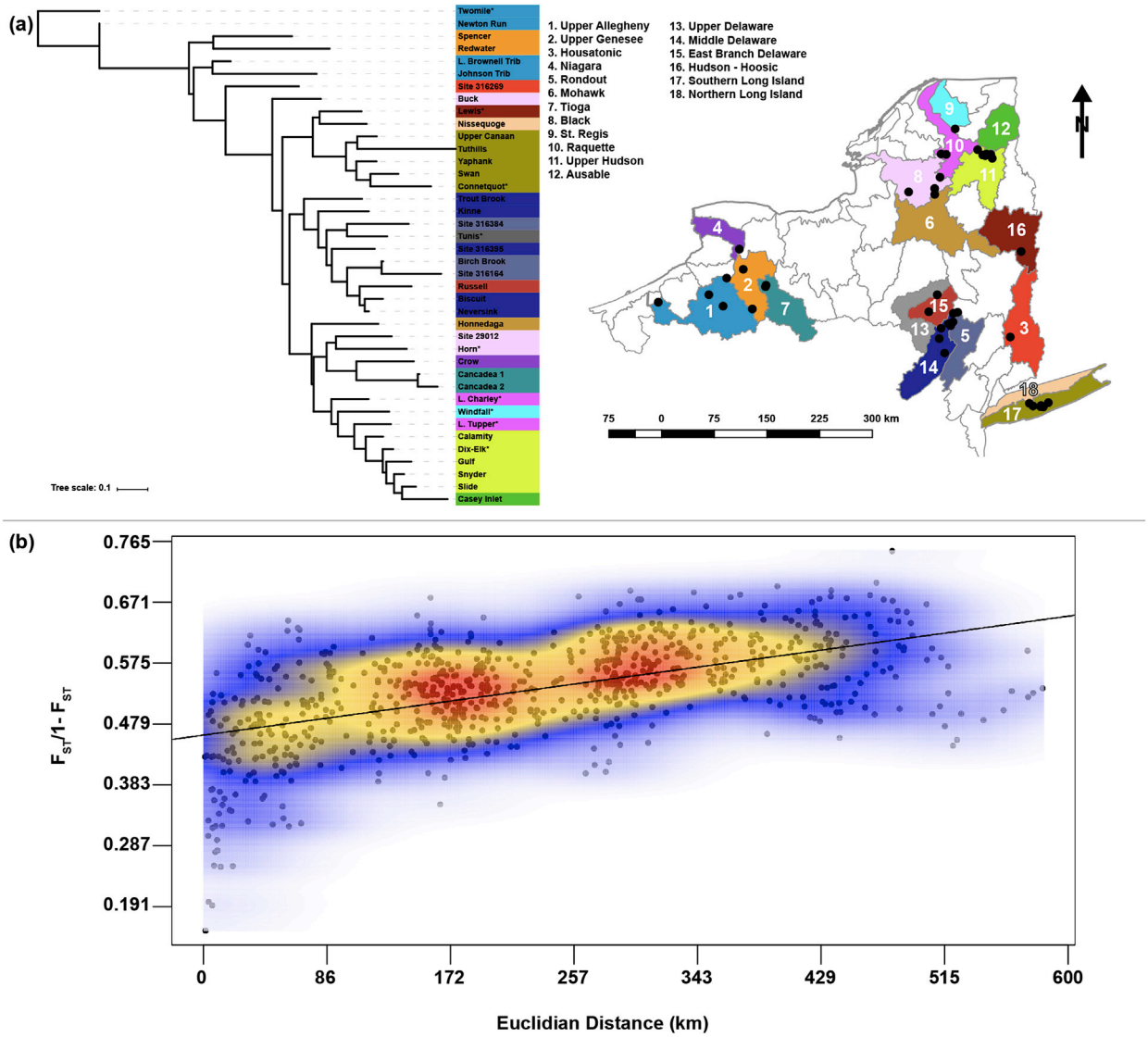

Fig. 3. Phylogram based on Nei's genetic distance (*D*; Supplemental 3) paired with New York State watershed map (a). Sampling locations include their watershed designation colored according to the map. Asterisks demarcate previously identified or confirmed native 'heritage' populations. Isolation-by-distance plot showing results of a Mantel test performed to identify the correlation between genetic differentiation and Euclidian distance between putatively native sampling locations (b). A positive correlation was found to be highly significant ($P < 0.0001$, $R^2 = 0.3941$). Color intensity reflects density of pairwise comparisons. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

assemblages worldwide (Ricciardi & Simberloff, 2009; Laikre et al., 2010). In the Northeastern United States, an extended history of habitat degradation, coupled with grave climate predictions, paints a disturbing picture, not only for the future of Brook Trout, but for all cold-water fish species. The putatively negative effects of translocations on aquatic gene pools, and the need for monitoring of such effects is not a new concept (Ryman, 1981) and has, in fact, been repeatedly stressed in the scientific literature (Hindar et al., 1991; Waples, 1999; Naish et al., 2007; Marie et al., 2010). Nevertheless, many government agencies have been slow to amend stocking practices and/or embrace genetic techniques to aid in conservation planning and native species recovery.

In this study, we integrated conventional techniques for microsatellite analysis into a novel framework for identifying native Brook Trout populations. Although long-term stocking has led to introgression from hatchery strains across many areas, we found that a large portion of Brook Trout populations throughout New York State still exhibit little to no sign of introgression from strains stocked over the past 3 decades. These populations are genetically distinct and exhibit significant genetic differentiation, as well as strong landscape-level genetic diversity based on sampling location. The patterns of genetic structure found in these populations are largely consistent with previously inferred native biogeographic structure based on postglacial recolonization patterns, which are dictated by their respective watershed boundaries and drainage basins. The

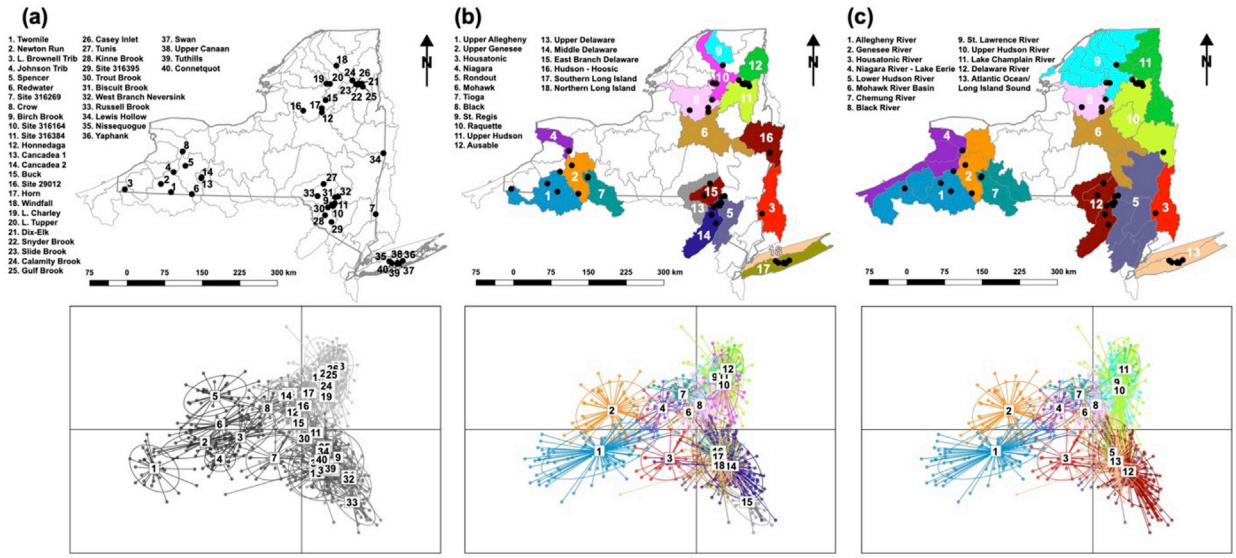

Fig. 4. Ordination plots of discriminant analysis of principal components (DAPC) comparing all putatively native samples included in the study with watershed geography at three hierarchal levels; the population level (a), the watershed level (b), and the major drainage basin level (c). The DAPC analysis was carried out retaining 50 principal components. Sampling locations with respect to each hierarchal level are labeled with corresponding colors and inertia ellipses in the DAPC, each dot represents the inferred population assignment for each individual fish. The watershed maps depict New York State, USA. Watershed and drainage basin boundaries are demarcated and colored according the DAPC plots. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

findings presented here offer insight into how native fish assemblages have been affected by long term stocking practices in the Northeastern United States and may also act as a practical guide for fisheries managers and researchers looking to incorporate watershed geography as a means to identify native populations of conservation priority elsewhere when reference samples for stocked strains are not always available.

When assessing the level of introgression exhibited across the state, it is apparent that although some amount of admixture is attributed to stocked fish across roughly half of the sampling locations examined, in most cases it is present in minimal amounts (<30%; Fig. 1). The majority of introgression recorded across sampled locations is largely concentrated within the populations of the Adirondack Park. Brook Trout fishing is a major tourist draw to the Adirondacks and a heavy contributor to the local economies there (Mullen and Menz, 1985). the State of New York therefore puts comparatively greater effort into enhancing its Brook Trout fishery for angling purposes in this region.

Introgression in other areas of the State was minimal in most cases, with the exception of Long Island, which exhibited heavy influence from the Cold Spring Hatchery strain at several sites, although two of these sites (Massapequa and Shu Swamp) were previously extirpated before stocking, suggesting that the hatchery fish in these areas do not hold a selective advantage over residents, given that there were simply none there. Six populations exhibited ancestry attributed entirely to stocked fish, while only nine populations exhibited ancestry attributed to stocked fish in the 50–99% ($Q_M = 0.50$ - 0.99; Fig. 1) range, a finding that stands in stark contrast to previous examinations of repeatedly stocked lakes and ponds in the state, which demonstrated extensive swamping out by stocked fish (Bruce et al., in preparation). This may suggest that stream-dwelling Brook Trout populations possess higher resiliency to stocked fish, or that stocked fish are less suited to hybridization in stream environments. Whatever the underlying cause of this phenomenon may be, these results partially confirm managers' expectations for a "put and take" fishery (at least outside of the Adirondack Park). If future management goals shift to increasing abundance through natural reproduction, stocking on top of wild stream-dwelling Brook Trout may be futile, given that the vast majority of ancestry at many of these locations is attributed to novel genetic structure rather than that of commonly stocked strains in the region.

Table 3
Hierarchical analysis of molecular variance between all samples with less than 5% ancestry attributed to stocked strains. Percentage of total variance (%) and *F*-statistics (F_{ST}, F_{IS}, F_{IT}) for each hierarchical level represented.

Comparison	Among groupings		Among individuals		Within individuals	
	%	F_{ST}	%	F_{IS}	%	F_{IT}
Population	29	0.291	0	0.006	71	0.295
Watershed	21	0.210	9	0.115	70	0.301
Major drainage	18	0.178	12	0.152	70	0.303

When comparing the putatively native populations to each other, we found that diversity measures varied widely among populations, and patterns seemed consistent with habitat constraints at individual locations rather than large-scale differences based on regions. Only two populations exhibited heterozygote deficiencies, while the majority of populations exhibited F_{IS} values not far from zero. Interestingly, no putatively native population exhibited a heterozygote excess. This provides some support for the idea that many of these populations are reproductively isolated; nevertheless, the general absence of heterozygote deficiencies suggests that if they are isolated, it is not a deterrent to within population gene flow. Effective population sizes varied widely and largely reflected diversity indices for most populations, although several outlier locations did produce markedly increased estimates compared to the majority of others, suggesting that fish from these sampling locations may belong to larger, interconnected populations. In addition, upper bounds of infinity suggest that some level of bias may be present in effective population size estimates across sample sites, although these measures still provide valuable information from a comparative standpoint.

The results of our demographic analyses, when taken in conjunction with the phylogram, and isolation by distance plot, all suggest local similarities in genetic structure, with differences between population groupings dictated by physical distance rather than changes in genetic structure associated with introgression or genetic drift (which would be expected to disrupt this pattern). The strength of these relationships in many cases was surprising, given the widespread historical stocking of non-local source strains in most of these areas. There were, however, several outliers in the analysis. Buck Creek, in particular, did not group with the other populations from the Black River watershed based on distance measures or the STRUCTURE plot, and diversity measures suggest that this population did not exhibit genetic structure consistent with a bottleneck effect. This may indicate that non-local stocked fish for which we do not have reference samples may have been released by a private hatchery and hybridized with, or become naturalized in, this population. Overall, genetic structure as exhibited by the STRUCTURE analyses and distance measures was largely what one would expect of native fish that have not been influenced by stocking practices.

Brook Trout genetic population structure is strongly influenced by watershed boundaries and drainage basin level boundaries across all populations examined in the study, as demonstrated by both DAPC plots and AMOVA results (Fig. 4, Table 3). Results of the AMOVA analyses suggest that roughly 20% of the variation exhibited across all samples can either be attributed to watershed or major drainage basin boundaries, while DAPC plots largely reflect the map of sampling locations at all hierarchical levels. These results strongly reflect groupings based on postglacial recolonization via oceanic routes and are consistent with past studies of native Brook Trout population genetic structure in the Northeastern United States (Perkins et al., 1993a, 1993b; Danzmann et al., 1998; Castric et al., 2001). When the results of our multiple analyses are considered collectively, they offer considerable evidence for the native origin of the majority of populations that showed little to no introgression from state directed stocking activities. Furthermore, with a handful of exceptions, only moderate to minor influence of hatchery stocks on wild Brook Trout population genetic structure was observed for the remaining populations found throughout New York State.

This study has demonstrated the utility of using genetic techniques to detect native populations across the landscape based on genetic biogeographic population structure. The strong population subdivision observed among putatively native fish contrasts with the spatial homogeneity among stocked populations demonstrated in previous work (Marie et al., 2010; Bruce et al., in preparation). In most cases, the pattern of differentiation across proximal sampling locations was comparatively low, as expected for native assemblages. Patterns of population structure were also congruent with prevailing postglacial recolonization hypotheses based on differences at both the watershed and drainage basin levels, confirming the physical basis underlying the acute levels of genetic variation exhibited across Brook Trout populations in New York State (Perkins et al., 1993a, 1993b). The populations that fit into this geographic framework, excluding rare outliers such as Buck, which seem likely to have experienced introgression from stocked fish for which we lack reference samples, should therefore be considered for the same level of protection as other 'heritage' strain Brook Trout present in the Northeastern United States in order to ensure their future viability and protect their unique genetic constitution.

Given the strong differences in genetic structure present at the population level and the sporadic nature of introgression across the landscape, determining strict conservation units based on geographic or watershed boundaries is problematic. Therefore, we argue for the prioritization of conservation at the level of individual populations, with native populations deserving the most attention, given their comparatively unique genetic structure, heightened likelihood of local adaptation, and their continued persistence despite widespread habitat degradation (Smith and Chesser, 1981; Fraser et al., 2011). Conservation prioritization of hybridized populations should follow, with elevated attention for populations that show minimal levels of admixture attributed to stocked fish, given that a number of studies have suggested when stocking ceases that hybridized populations have the potential to return to their natural genetic constitution (Perrier et al., 2013; Valiquette et al., 2014; Létourneau et al., 2018). Wild populations that owe their ancestry entirely to stocked fish should be prioritized last, given their domesticated origins and genetic ambiguity. Finally, for the many wild brook trout populations whose origins remain undetermined, the management paradigm in New York State should begin with the presumption of native origin. This will help to ensure the preservation of unique genetic structure across the species' native range, and maintain diversity at levels that may facilitate adaptation across changing landscapes. The results presented here also suggest that by combining traditional admixture analysis with watershed geography, researchers may be able to identify other native populations, even when reference samples for historically stocked or translocated species may not be available. Therefore, this combined approach to the identification of native species based on genetic and hydro-geographic structure may be applied by managers

and researchers elsewhere to confirm native assemblages of other genetically structured fresh-water aquatic species for conservation or management purposes.

Given the level at which humans have extensively altered the global environment by transforming local geography and enhancing the mobility of non-native biota, conserving diversity at all organizational levels, from genetic diversity within populations to the diversity of ecosystems across landscapes, is essential to the continued livelihood of all species. In the absence of major adaptations to policy, the ongoing influence of anthropogenic disturbances will continue to have a pronounced effect on biodiversity across the globe (Chapin et al., 2000). The well-established techniques presented here have been shown to aid in the discovery of previously unidentified genetic diversity across large geographic scales, while helping to unravel patterns associated with introgressive hybridization and natural genetic structure. The native populations examined in this study represent an irreplaceable part of New York State's natural heritage, and it is our hope that this work will, at least in part, contribute to their longevity for many generations to come.

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Conflict of interest

The authors declare that there is no conflict of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.gecco.2019.e00682.

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